

RESEARCH ARTICLE

Automatic Detection and Classification of Dental Anomalies and Tooth Types Using Transformer-Based Yolo With GA Optimization

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Internal Review Committee.

ABSTRACT Dental pathology detection and tooth type categorization are crucial yet often error-prone and time-consuming tasks in dental diagnostics. This paper presents a novel approach using Transformer-Based You Only Look Once (YOLO) for automating dental pathology detection and tooth type categorization from panoramic X-ray images. Manual dental examinations are error-prone and time consuming. Therefore automated systems are required. Preprocessing techniques such as image resizing, rotation, Gaussian smoothing, and histogram scaling are used to enhance image quality and analysis. A significant enhancement is the integration of a multi-head self-attention block into the Cross Stage Partial Darknet (CSPDarknet) backbone within the YOLOv4 architecture. This addition enabled the model to capture comprehensive contextual information and learn distinct feature representations from the dental images. Inspired by Transformer models, the multi-head self-attention mechanism improves the capacity of the model to intricate spatial relationships in images, enhancing detection and classification performance. In addition, the system employs Genetic Algorithm (GA) optimization for hyperparameter tuning, which fine-tunes model parameters to maximize performance metrics and enhance accuracy and robustness in dental diagnostics. The system achieved an average precision, recall rate of 99.5% and F1 score of 99.41% and a detection and classification accuracy of 99.31%. These results highlight the effectiveness of advanced preprocessing, deep learning enhancements, and GA optimization for precise dental pathology detection and tooth type categorization using panoramic radiography.

INDEX TERMS CSP-Darknet, image preprocessing, multi-head self-attention, panoramic X-ray, Transformer-based YOLO.

I. INTRODUCTION

Recent breakthroughs in deep learning and computer vision have brought about significant changes in numerous industries, notably healthcare and dentistry. These advancements have become crucial for automating and enhancing the analysis of medical images through precise object categorization and detection [1]. The application of deep learning extends across a range of fields, including diabetes diagnosis [2],

Kidney Tumor Classification [3], Urinary Stone Segmentation [4], precise image classification [5] and geriatric care [6]. In dental image processing, the application of deep learning holds considerable promise for improving the diagnostic accuracy and treatment planning, allowing complex tasks such as tooth recognition, categorization, and segmentation to be executed with unprecedented precision [7], [8]. Despite these advancements, several persistent challenges have hindered widespread adoption of deep learning in dental healthcare. Ensuring high accuracy is paramount, as errors can lead to misdiagnoses, which is critical given the complexity of

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FIGURE 1. An Orthopantomogram X-ray.

dental images characterized by varying contrast, cluttered backgrounds, and artifacts. Real-time processing is essential for clinical use. However it is challenging to achieve this without compromising the accuracy. Furthermore, integrating these advanced models with the existing dental systems presents significant difficulties, and many models struggle with generalizability across diverse populations and imaging conditions. The lack of transparency nature of deep learning models complicates the interpretability of decisions for clinicians. Moreover, ensuring the robustness of the models across different patient demographics and imaging modalities necessitates rigorous validation and comprehensive data augmentation strategies. Ongoing research and innovation are crucial for addressing these challenges. The next section reviews some notable studies in this field.

The remainder of this paper is organized as follows. Section II describes the literature survey, Section III highlights the contribution of the work, and Section IV presents the proposed methodology. The results and discussion are presented in Section V, and Section VI concludes the paper.

II. LITERATURE SURVEY

Recent advancements in dental image analysis have explored a variety of techniques to enhance accuracy and address challenges related to image variability and artifacts. Traditional methods such as integral projection and affine transformation have been applied in a modified AlexNet CNN for tooth detection, achieving commendable accuracy despite encountering misalignment issues in neighboring teeth [9]. Similarly, manual region-of-interest (ROI) placement combined with AlexNet-based classification demonstrated notable accuracy in dental CT images. However, preprocessing variability and metal artifacts remained significant challenges, affecting the model's robustness [10].

To further improve automated tooth classification and segmentation, deep learning techniques have been extensively utilized, leveraging both 2D and 3D data representations for enhanced performance. Early approaches employed Artificial Neural Networks (ANNs) for classifying 3D tooth cusps, achieving high accuracy. However, these models

exhibited a strong dependency on training dataset size and were prone to misclassification errors, limiting their generalizability in diverse clinical scenarios [11]. In parallel, deep learning has been applied to cone-beam CT imaging, where CNNs incorporating global average pooling achieved classification sensitivities of 88% for incisors, 86% for canines, 84% for premolars, and 90% for molars. However, the model's performance was significantly influenced by image noise, which posed challenges in real-world clinical applications [12].

With the increasing adoption of 3D data-based models, researchers have explored more sophisticated approaches to improve classification and segmentation. Automated segmentation methods integrating neural networks with shape models have achieved precision and recall values of 0.790 and 0.827, respectively, when wisdom teeth were excluded. Despite these promising results, these models face challenges in processing unfamiliar regions, limiting their ability to generalize across diverse dental structures [13]. Further advancements in 3D convolutional neural networks (CNNs) have significantly improved tooth identification, with one model achieving 95.8% precision and 96.1% recall. However, these models require extensive datasets for optimal performance, emphasizing the necessity of robust data collection strategies to ensure generalizability [14]. Similarly, a deep CNN applied to 3D dental models demonstrated impressive mesh labeling accuracy, though its applicability remained confined to dental mesh datasets [15]. In an effort to further enhance classification accuracy, Sparse Voxel Octree (SVO) integration with CNNs has been employed for 3D dental model classification, achieving accuracies of 95.96% and 88.06%. However, hierarchical segmentation constraints and limited dataset sizes continue to hinder their generalizability across varied dental conditions [16].

Recent research has also explored innovative deep learning architectures for improving tooth classification and anomaly detection. A Faster R-CNN model applied to Orthopantomograms (OPG) radiographs achieved 90% detection accuracy and 99% classification accuracy, effectively identifying two tooth types and five dental irregularities [17]. Furthermore, a CNN-based automated diagnostic system demonstrated 97.2% accuracy in classifying teeth (molars, premolars, canines, and incisors) and detecting anomalies such as cavities and fractures, providing valuable insights into the strengths and limitations of CNN-based approaches [18].

Hybrid and ensemble-based models have also shown promise in dental image analysis. A dual-stream feature extraction framework combining the Swin Transformer and MobileNetV2 was integrated with robust preprocessing techniques and a bagging ensemble classifier, achieving 95.6% accuracy and a 95.5% Dice similarity coefficient. This model outperformed state-of-the-art methods in dental disease classification, highlighting its potential for clinical applications [19]. Similarly, a comprehensive automated system for analyzing OPGs was developed, incorporating three modules: tooth detection and counting using a U-Net refined

with YOLOv3, abnormality classification using InceptionV3 transfer learning, and detection of treated pathologies (e.g., root canals, crowns, and implants) using Faster R-CNN and InceptionV3. Despite its promising results, the system exhibited moderate to fair accuracy, indicating the need for further refinement [20]. In the realm of panoramic dental X-ray analysis, deep learning models have been employed to enhance classification accuracy. A deep convolutional neural network incorporating three convolutional layers and three fully connected layers was developed to classify canine, incisor, molar, and premolar tooth types, achieving 95% accuracy. However, the model was susceptible to misclassification of adjacent teeth, underscoring the need for more precise localization techniques [21]. These studies collectively underscore the importance of robust preprocessing techniques, comprehensive validation, and optimization strategies to mitigate artifacts and enhance the performance of dental image analysis.

Building on these insights, our proposed work aims to overcome the aforementioned challenges by introducing an automated system that leverages the YOLOv4 object detection framework. This system aims to detect and classify four specific tooth types (canines, incisors, molars, and premolars) and two anomalies (capped and impacted teeth) from panoramic X-ray images, commonly referred to as Orthopantomograms (OPG) (Fig. 1). YOLOv4 was selected for its outstanding real-time object detection capabilities, superior accuracy, efficiency and is characterized by its advanced architecture, including robust feature pyramid networks and attention mechanisms that enhance detection in complex dental images. Despite advancements in newer versions, YOLOv4's robustness and extensive research support justify its selection for this study.

III. CONTRIBUTION

The proposed approach identifies and classifies four distinct tooth types namely the Molar, Premolar, Incisor, and Canine and detects dental anomalies such as impacted teeth and caps from panoramic radiographs. The key contributions of this method are as follows:

- 1) The Darknet backbone of YOLOv4 was modified by incorporating a multi-head self-attention layer, which enhanced feature extraction and global context capture in dental images.
- 2) A genetic algorithm (GA) was utilized to systematically optimize the hyperparameters, improve the model performance and reduce the need for extensive manual annotation.
- 3) The system demonstrated high precision, recall, and F1 scores, indicating its effective detection and classification capability.
- 4) The developed system may offer real-time processing and ease of integration, reduces errors, and streamlines the diagnostic process for dental healthcare professionals.

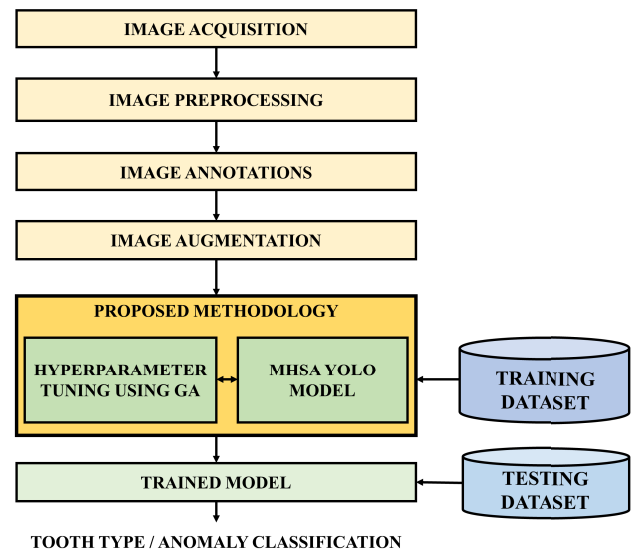


FIGURE 2. Block diagram of the proposed system.

IV. PROPOSED METHODOLOGY

Our proposed methodology aims to automate the detection and categorization of four distinct tooth types including molars, premolars, incisors, and canines, along with the identification of dental anomalies namely impacted teeth and those bearing a dental crown, commonly referred to as a cap from panoramic radiographs. By utilizing image acquisition, preprocessing techniques, and image annotations, the developed system demonstrates potential for enhancing dental diagnostic processes and improving patient care. A block diagram of the proposed system is shown in Fig. 2.

A. IMAGE ACQUISITION

Our initial phase involved the acquisition of panoramic radiographs. The radiograph captures expansive radiographic views of the entire dental arch, offering a two-dimensional portrayal of spannings from one ear to another. Unlike traditional X-rays, panoramic radiography employs focal plane tomography to integrate images from multiple planes, resulting in a singular panoramic image. This technique ensures that the upper (maxilla) and lower (mandible) jaws are clearly depicted within a focused area known as the focal trough.

In our study, we collected and analyzed a total of 475 panoramic images, which were used to identify 15,000 individual teeth belonging to various tooth and anomaly classes.

B. IMAGE PREPROCESSING

Image pre-processing is an essential step in preparing data for machine learning models, as it ensures consistency and highlights the features necessary for accurate training. For our study, we followed a systematic approach to optimize the dataset. The process began with resizing all

the images to a standard size, ensuring uniformity and compatibility with the model's architecture. This helped prevent issues caused by inconsistent image dimensions. We then applied noise reduction techniques, such as median filtering, to remove artifacts and random pixel intensity variations. This step improved the clarity of the images while preserving important details. To ensure proper alignment, we adjusted the orientation and size of the images using rotation and scaling techniques, all while maintaining their aspect ratios to preserve spatial accuracy. Next, cropping was used to focus on the regions of interest (ROIs) and remove any unnecessary parts of the images, making the data cleaner and reducing computational load. Finally, we applied smoothing techniques, such as a median filter, to reduce abrupt changes in pixel intensity, creating a more polished and uniform appearance. Together, these steps ensured that the dataset was consistent, high-quality, and ready for training. By minimizing noise, aligning images properly, and emphasizing critical features, this preprocessing workflow allowed our model to learn intricate patterns more effectively, ultimately improving its overall performance in the task. The preprocessing steps are illustrated in Fig.3.

C. IMAGE ANNOTATIONS

The third step involved image annotation, where we define precise bounding boxes around the different types of teeth in the images. These annotations were created using Roboflow [22] to generate ground-truth bounding boxes. A total of 15,000 diverse tooth categories, comprising molars, premolars, canines, incisors, impacted teeth, and caps, were meticulously annotated from the acquired OPG radiographs, and annotations can be obtained in multiple file formats. In accordance with our needs, we obtained annotations in the YOLOv4 format for integration with our model.

D. IMAGE AUGMENTATION

The fourth stage comprises image augmentation, which encompasses a range of operations including rotation, flipping, cropping, and shearing. The resulting dataset, which consisted of 20,000 teeth, was subsequently divided into two subsets: 90% for training and 10% for testing.

E. TRAINING

The fifth stage involved training the proposed attention-based YOLOv4 model which consisted of a self-attention mechanism and convolution layers of YOLOv4. Panoramic radiographs, coupled with their corresponding annotations, served as the inputs for the model. Through this training process, the model acquires distinctive features of various types of teeth by utilizing objects within the ground truth bounding boxes as a reference.

As a state-of-the-art object detection model, YOLOv4 operates in a single stage, simultaneously performing both object localization and classification [23]. The architecture comprises three key components. The first component, the

backbone, leverages a novel combination of Multi-Head Self-Attention (MHSA) and Darknet architecture to extract more differentiated features from the input image for further use. Following the feature extraction in the backbone, YOLOv4 utilizes a neck component comprising a Path Aggregation Network (PANet). The function of PANet is to strategically combine features extracted from different stages of the backbone, ensuring that they align with the requirements of various detection stages. Finally, the architecture reached its head component. YOLO detection heads, employed at different scales, predict the bounding boxes. These heads are responsible for the crucial tasks of both object localization and classification. Notably, YOLOv4 delivers significant performance improvements compared to previous versions. Fig.4 illustrates the architecture of the Proposed Attention-based YOLOv4 Architecture along with CSP block used in our work.

1) BACKBONE OF YOLOV4

The YOLOv4 model uses CSPDarknet53 as its backbone, which consists of 53 convolutional layers (3×3 and 1×1) for the feature extraction. This backbone employs a cross stage partial (CSP) architecture, which splits layers into two paths—one processed through convolutions and the other bypassing them—before merging. This enhances feature learning compared to YOLOv3 Darknet. Additionally, YOLOv4 integrates a Multi-Head Self-Attention (MHSA) block into CSPDarknet, further improving the model's capability for diverse and superior feature learning by enabling global self-attention across feature maps.

MHSA-Darknet integrates Multi-Head Self-Attention (MHSA) into Darknet for YOLO, enabling global self-attention across 2D feature maps. YOLOv4 comprises seven stages (P1-P7) with strides [2, 4, 8, 16, 32, 64, 128], each stage containing multiple CSPDark blocks, specifically [1, 3, 15, 15, 7, 7, 7] for P7. In MHSA-Darknet, transformer layers are strategically applied at stage P7, avoiding earlier stages (P3-P6) to prevent the loss of contextual information in large feature maps. P7 typically includes seven CSP bottleneck blocks with 1×1 and 3×3 convolutions. Incorporating Multi-Head Self-Attention (MHSA) enhances feature extraction, allowing a dynamic focus on relevant input features. This improves detection accuracy and robustness and extends applications to real-time object detection, image segmentation, and classification. In dental imaging, MHSA-Darknet's advanced neural architecture, with MHSA mechanisms, enhances the diagnostic accuracy and efficiency in classifying dental tooth types using OPG X-ray images.

2) NECK OF YOLOV4

YOLOv4 integrates Spatial Pyramid Pooling (SPP) and a Path Aggregation Network (PAN) to gather and merge feature maps from different backbone stages. It acts as a crucial intermediary between the backbone and the network head, enhancing feature propagation through PANet. This architectural framework facilitates effective information flow

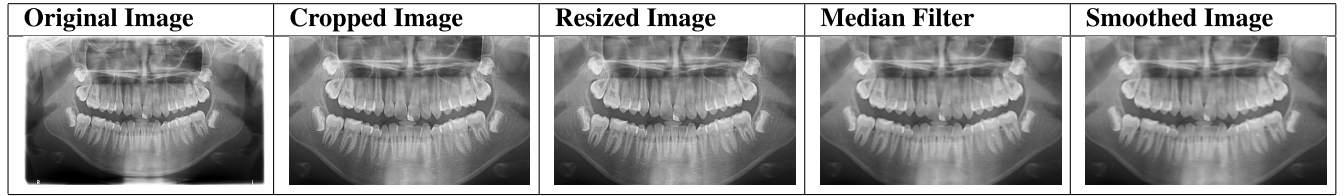


FIGURE 3. Image preprocessing steps.

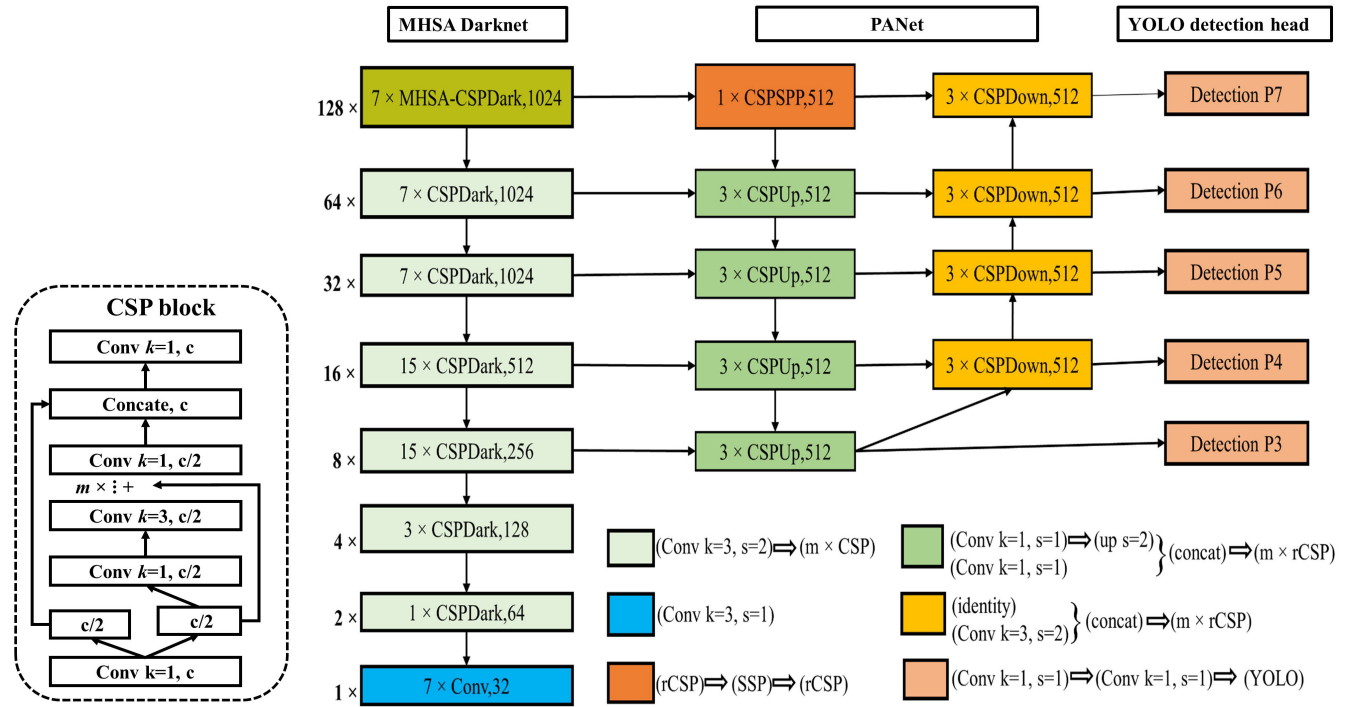


FIGURE 4. Proposed attention-based Yolov4 architecture along with CSP block.

by concatenating current inputs with data from the preceding layers and optimizing feature integration for subsequent processing in the network.

The neck in YOLOv4 integrates Spatial Pyramid Pooling (SPP), which uses multiple spatial bins to produce fixed-size feature maps, thereby ensuring robustness to object deformation and enhancing the handling of diverse object sizes. PANet improves upon traditional Feature Pyramid Networks (FPNs) by incorporating adaptive feature pooling and bottom-up path augmentation for better multi-scale object detection. This dual-path approach contrasts with the unidirectional flow of FPNs by optimizing feature integration and propagation across network levels [24], [25].

PANet aggregates multi-scale features through a transformation that effectively combines input features from levels 3 to 7, denoted as $\tilde{P}^{in} = (P_3^{in}, \dots, P_7^{in})$. Each P_i^{in} corresponds to a feature level with a resolution of $\frac{1}{2^i}$ of the original image resolution, such as P_3^{in} with a resolution of 128×128 for an

input resolution of 1024×1024 , and P_7^{in} with a resolution of 8×8 . The aim is to identify a transformation f that effectively combines these input features with intermediate features $\tilde{P}^{td} = (P_1^{td}, P_2^{td}, \dots)$ to produce a new set of output features $\tilde{P}^{out} = f(\tilde{P}^{in})$. The intermediate and output features in layer 6 are as follows:

$$\begin{aligned} P_6^{td} &= \text{Conv}(P_6^{in} + \text{Resize}(P_7^{td})) \\ P_6^{out} &= \text{Conv}(P_6^{td} + \text{Resize}(P_5^{out})) \end{aligned} \quad (1)$$

where, P_6^{in} refers to the input feature at layer 6, P_6^{td} and P_7^{td} refers to the intermediate feature at layer 6 and 7, P_5^{out} and P_6^{out} refers to the output feature at layer 5 and 6 respectively, Resize refers to adjusting the resolution through either upsampling or downsampling to match the dimensions and Conv refers to the convolutional operations used for processing features.

3) HEAD OF YOLOV4

The primary role of the head component is to determine the coordinates of a bounding box and to classify the object contained within that specific bounding box. In YOLOv4, the structure of the head closely resembles that of its predecessor, YOLOv3. The network head processes the combined features, producing predictions for bounding boxes, object confidence scores, and classification probabilities. Both YOLOv3 and YOLOv4 use one-stage object detectors in their detection heads.

The system predicts the four coordinates (a_x, a_y, a_w, a_h) for the bounding boxes. Let (c_x, c_y) denotes the coordinates of the top left corner of the bounding box and (w_{gt}, h_{gt}) and (w, h) denote the weight and height of the ground truth bounding box and predicted bounding box, respectively. The predictions are formulated as follows [26]

$$\begin{aligned} b_x &= \sigma(a_x) + c_x \\ b_y &= \sigma(a_y) + c_y \\ w &= w_{gt} e^{a_w} \\ h &= h_{gt} e^{a_h} \end{aligned} \quad (2)$$

where (b_x, b_y) denotes the center coordinates of the predicted box, $\sigma(a_x)$ and $\sigma(a_y)$ are functions dependent on a_x and a_y respectively.

YOLOv4 focuses on enhancing performance through innovative features. It introduces input image variation to boost learning from diverse datasets and enhances training with advanced data augmentation techniques such as photometric and geometric distortions, MixUp, CutMix, and Mosaic for small objects. Additionally, it integrates the CIOU Loss function for precise bounding boxes, Cross-mini Batch Normalization for smaller datasets, dropblock regularization to prevent overfitting, and the Mish activation function to improve the accuracy and stability [23].

CIOU is an object detection metric that improves upon Intersection over Union (IoU) by considering additional geometric insights for a more accurate bounding box evaluation. CIOU Loss is given by:

$$L_{CIOU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (3)$$

where IoU is the Intersection over Union between the predicted and ground truth bounding boxes. $\rho(b, b^{gt})$ is the Euclidean distance between the center points of the predicted bounding box b and the ground truth bounding b^{gt} . c is the diagonal length of the smallest enclosing box that can encompass both the predicted and ground truth bounding boxes. α is a positive trade-off parameter defined as:

$$\alpha = \frac{v}{(1 - IoU) + v} \quad (4)$$

v is a measure of the aspect ratio consistency between the predicted and ground truth bounding boxes and is calculated as follows:

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w_{gt}}{h_{gt}} - \arctan \frac{w}{h} \right)^2 \quad (5)$$

where w and h are the width and height of the predicted bounding box, w^{gt} and h^{gt} are the width and height of the ground truth bounding box respectively.

Cross Mini-Batch Normalization (CMBN) is an extension of traditional batch normalization which leverages statistics computed from multiple mini-batches to improve the stability and performance of deep neural networks. In the CMBN, the mean and variance are calculated across multiple mini-batches, providing more stable batch statistics. The inputs were normalized using these aggregated statistics to ensure zero mean and unit variance, which aids in faster convergence and robust training. The mean and variance are given by:

$$\begin{aligned} \mu &= \frac{1}{M} \sum_{m=1}^M \mu_m \\ \sigma^2 &= \frac{1}{M} \sum_{m=1}^M \left(\sigma_m^2 + (\mu_m - \mu)^2 \right) \end{aligned} \quad (6)$$

where M is the number of mini-batches. μ and σ_m^2 are the mean and variance of the m -th mini-batch respectively.

CMBN is given by:

$$\hat{x}_i^{(m)} = \frac{x_i^{(m)} - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (7)$$

where $x_i^{(m)}$ is the i -th input of the m -th mini-batch. ϵ is a small constant added for numerical stability.

4) HYPERPARAMETER TUNING USING GA

Hyperparameter tuning in the YOLOv4 framework involves optimizing a range of configurations to refine the performance of the teeth detection algorithm. YOLOv4, which is known for its speed and accuracy, allows fine-tuning of various parameters to improve the detection performance on particular tasks or datasets. YOLOv4 hyperparameter adjustment can be determined as follows:

Learning Rate: The learning rate determines the step size during each iteration while minimizing the loss function. Higher rates can lead to faster convergence but risk overshooting optimal solutions. Conversely, lower rates converge more gradually but may yield better results.

Batch Size: The number of training samples processed in each iteration is determined by the batch size. Larger batches can improve generalization and accelerate convergence, however they require more memory. Smaller batches, on the other hand, might converge more slowly but might produce better generalization. Investigating different batch sizes is necessary to find a compromise between the model performance and training speed.

Anchor Boxes: These boxes are essential for predicting teeth bounding boxes, and have a major impact on the detection performance. Changing the quantity and aspect ratios can have an impact on the detection accuracy.

Number of Training Epochs: When training machine learning models, monitoring the training and validation loss, as well as other relevant variables, is critical for determining

the best termination point. The aim is to establish a balance between convergence and overfitting. This precise balance guarantees that the model generalizes successfully to new data while avoiding over-adaptation to the training set.

Number of filters: Filters in the convolutional layers determine the depth of the feature maps, and the capture essential patterns for teeth detection. Lower layers typically have fewer filters (e.g., 32, 64), whereas higher layers use more filters (e.g., 128, 256), balancing feature complexity and computational efficiency.

No. of Convolutional Layers: YOLOv4's feature extraction and computational complexity are greatly influenced by the number of convolutional layers in the model. While adding layers improves pattern recognition, it also increases the computing demands and the risk of overfitting. Using methods such as grid search and Bayesian optimization, the careful tuning of this parameter optimizes the framework for precise tooth detection.

A genetic algorithm (GA) is a search technique inspired by the principles of natural selection. It is commonly employed to find effective solutions for optimization and search challenges [27]. The fundamental steps in a GA are as follows:

- 1) Initially, the Genetic Algorithm (GA) creates a random population of bit strings, where each string symbolizes a potential solution for optimizing the hyperparameters. For our application, this population consists of 60 bit strings, each 25 bits long. Each string encodes specific hyperparameters: the learning rate is represented by a 2-bit binary substring, batch size by a 7-bit binary substring, number of anchors by a 3-bit binary substring, number of filters by a 7-bit binary substring, and number of convolutional layers by a 6-bit binary substring.
- 2) **Fitness Definition:** Fitness is defined as the measure of a solution's effectiveness in optimizing the model's hyperparameters, quantified by the model's performance metrics.
- 3) **Evaluation:** For each bit string, the five substrings—representing the learning rate, batch size, number of anchor boxes, number of filters, and number of convolutional layers—are converted to their respective numeric values. These values were then applied during the training and validation/testing processes of the system to determine the performance of the model. The fitness or accuracy of each bit string is computed based on this performance.
- 4) **Selection:** The selection process identifies bit strings with fitness or accuracy exceeding a predefined threshold. Only high-performance bit strings were selected for further operations.
- 5) **Reproduction:** The population evolves via three mechanisms: selection, crossover, and mutation. This repeated process of evaluation, selection, and reproduction occurs across multiple generations. The goal is to continuously improve the population by retaining

and refining the bit strings with fitness values above the specified benchmark. The integration of the Genetic Algorithm ensures that the model consistently achieves optimal performance, effectively tuning the hyperparameters for maximum efficiency.

V. RESULTS AND DISCUSSIONS

Our study employed the proposed model configured with 104 batches, 36 convolutional layers and a learning rate of 0.002.

A. GA OPTIMIZATION FOR HYPERPARAMETER TUNING

Table 1 presents an extensive overview of the genetic algorithm (GA) optimization utilized for hyperparameter tuning across various generations. The binary strings within the table encode the optimal configurations for the key parameters, including Learning Rate, Batch Size, Number of Anchor Boxes, Number of Filters, and Number of Convolutional Layers. In the initial row, which corresponds to the 50th generation, the binary string "10 1000000 101 1000000 110000" represents the highest fitness/accuracy configuration. This configuration breakdown revealed that the first substring corresponds to a Learning Rate of 0.005, the second substring indicates a Batch Size of 64, the third substring denotes 5 Anchor Boxes, the fourth substring reflects 64 filters, and the last substring signifies 24 Convolutional Layers. Conversely, the final row, representing the 1000th generation, showcases the binary string "00 1101000 101 1000000 100100" which achieves the highest fitness/accuracy of 99%. In this configuration, the first substring corresponds to a Learning Rate of 0.02, the second substring indicates a Batch Size of 104, the third substring denotes 5 Anchor Boxes, the fourth substring reflects 64 filters, and the last substring represents 36 Convolutional Layers.

Table 1 highlights the highest fitness/accuracy values attained, emphasizing the effectiveness of the genetic algorithm in precisely refining the model. This optimization process plays a pivotal role in improving the overall performance and precision of the model, and is seamlessly aligned with the objectives of advanced hyperparameter tuning methodologies in machine learning.

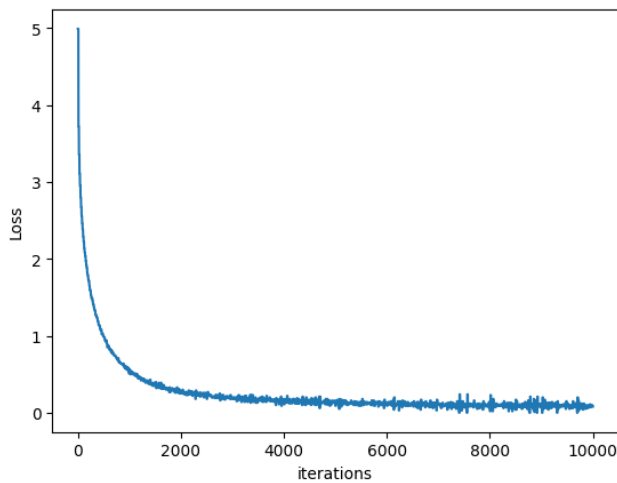
B. TRAINING LOSS

Although YOLO predicts multiple bounding boxes per grid cell, only one should represent a true positive object. To address this, the model selects the box with the highest Intersection over Union (IoU) with the ground truth. This strategy encourages specialization among predictions, where each box becomes more adept at detecting objects of specific sizes and aspect ratios.

Fig. 5 illustrates the relationship between iteration and loss, with the loss initially high and gradually decreasing as the training progresses. The model demonstrated the ability to achieve low loss, and an average loss of approximately 0.0437 was achieved.

TABLE 1. Hyperparameter optimization using GA.

Generations	Learning Rate	Batch Sizes (decimal)	No. of Anchors	No. of Filters	No. of Convolutional Layers	Highest Fitness/Accuracy
50	10 (0.005)	1000000 (64)	101 (5)	1000000 (64)	110000 (24)	80%
150	00 (0.002)	1110000 (112)	011 (3)	1111111 (127)	100000 (32)	85%
250	10 (0.005)	1111111 (127)	100 (4)	1000000 (64)	111111 (63)	88%
350	11 (0.008)	1010101 (85)	010 (2)	1000000 (64)	111000 (56)	90%
450	11 (0.008)	0111111 (63)	110 (6)	1000000 (64)	101000 (40)	92%
550	00 (0.002)	1110000 (112)	111 (7)	1000000 (64)	000101 (5)	93%
650	00 (0.002)	1000000 (64)	010 (2)	1000000 (64)	100000 (32)	94%
750	11 (0.008)	1111111 (127)	010 (2)	1111111 (127)	10000 (16)	95%
850	11 (0.008)	0100000 (32)	100 (4)	1001001 (73)	101100 (44)	96%
950	00 (0.002)	1010000 (80)	011 (3)	1111111 (127)	110000 (48)	97%
1000	00 (0.002)	1101000 (104)	101 (5)	1000000 (64)	100100 (36)	99%

**FIGURE 5.** Graph between training loss and iterations.

C. TESTING RESULTS

After the model was successfully trained, we examined how well it identified and categorized teeth in the raw testing dataset. The results, depicted in Fig.6, demonstrate the exceptional capability of the system in detecting essential dental structures such as molars, premolars, and caps. Furthermore, Fig.7 further illustrates the effectiveness of the model in detecting and classifying molars, premolars, incisors, and canines. Additionally, Fig.8 offers further insights by demonstrating the model's ability to detect caps alongside molars, premolars, and incisors. These combined findings underscore the robustness and reliability of the model in accurately detecting and categorizing various teeth within a dataset.

D. PERFORMANCE EVALUATION

The following metrics given below are used for evaluation of the dental detection and classification:

• Precision

Precision is a measurement that calculates how accurate our predictions are. It measures how many of the predictions that our model made are actually correct.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

where TP stands for True Positive and FP stands for False Positive respectively.

• Recall

Recall is a measurement that calculates how well our model finds all the positives.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

where TP stands for True Positive and FN stands for False Negative respectively.

• F1 Score

The F1-score is a measurement of a testing data's accuracy and is the harmonic mean of the precision and recall values. The equation for F1-score is given below.

$$\text{F1 Score} = \frac{2 \cdot \text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} \quad (10)$$

• Average Precision

Average precision is a metric that measures the precision of a model at various recall levels, providing an aggregate measure of performance across all thresholds. Average Precision (AP) is used to calculate the average precision value for recall value over 0 to 1.

$$\text{AP} = \int_0^1 p(r) dr \quad (11)$$

• Intersection over Union

Intersection over Union (IoU) is a metric used to evaluate the accuracy of an object detection model by comparing the predicted bounding box with the ground truth. Intersection over Union is used to describe the extent of overlap of two boxes.

$$\text{IoU} = \frac{A1}{A2} \quad (12)$$

where, A1 is the overlapping areas between predicted and ground truth bounding boxes and A2 is the union area of 2 boxes.

• Mean Average Precision

Mean Average Precision (mAP) evaluates an object detection model's performance by averaging precision

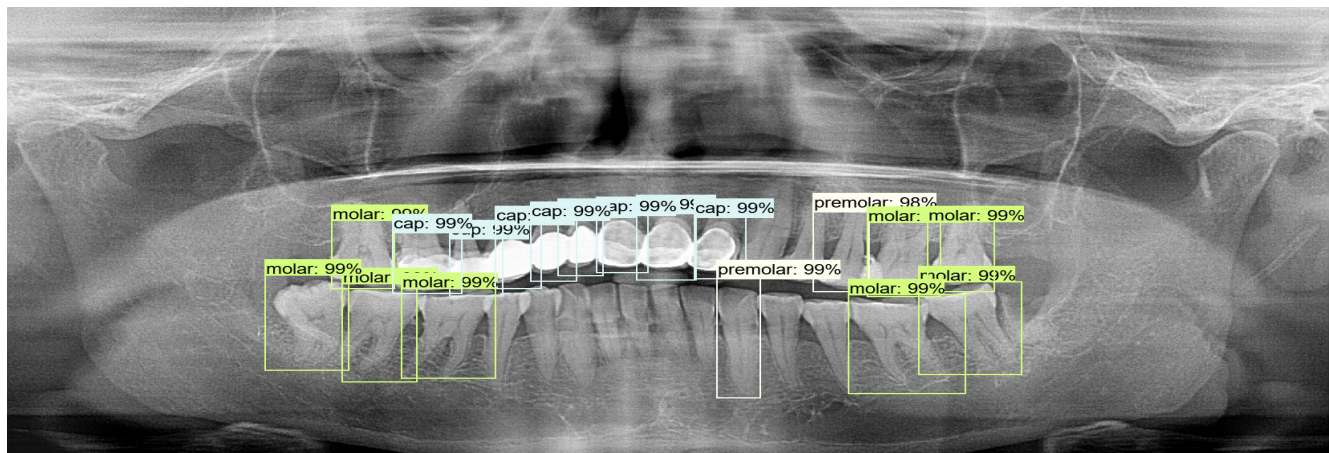


FIGURE 6. Output of detection and classification for test image-1.

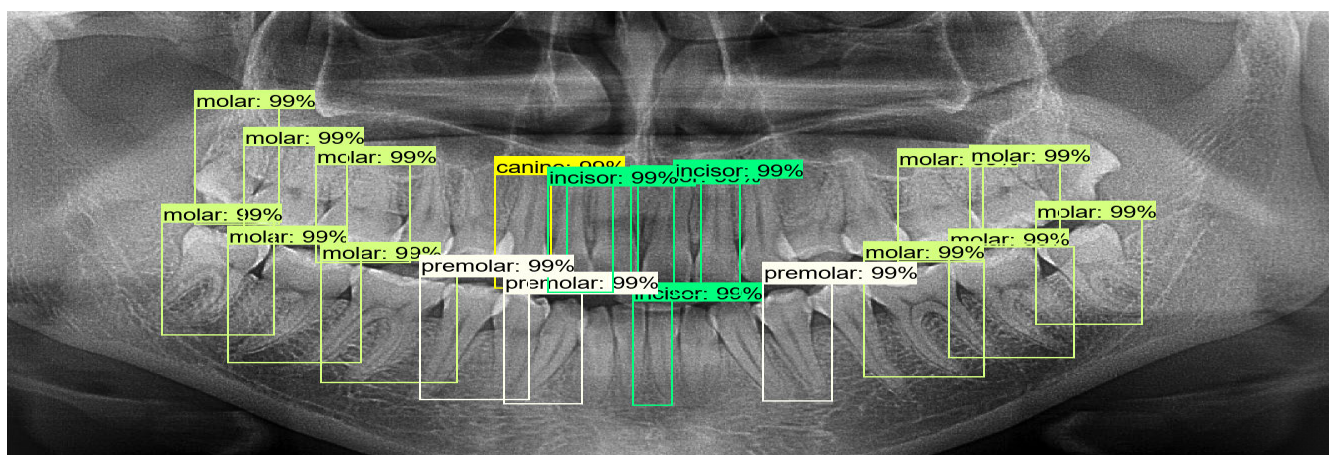


FIGURE 7. Output of detection and classification for test image-2.

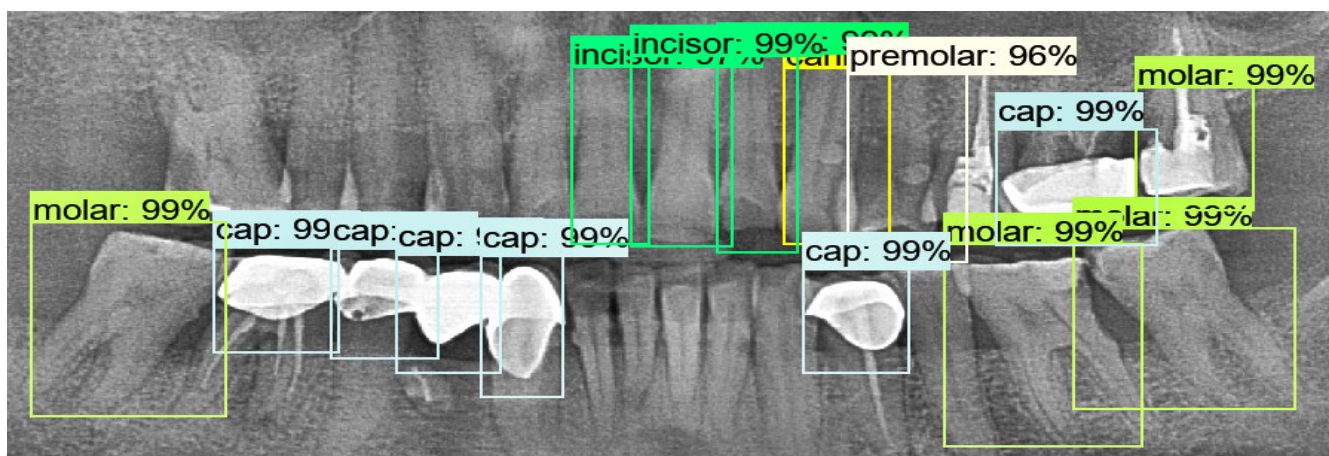


FIGURE 8. Output of detection and classification for test image-3.

scores at different recall thresholds for all classes.

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (13)$$

where, N represents the total number of classes, and AP_i denotes the Average Precision for each individual class.

TABLE 2. Evaluation metrics.

Category	Precision	Recall	F1 Score	Accuracy
Incisor	0.9969	0.9969	0.9969	0.9969
Canine	0.9901	0.9933	0.9917	0.9933
Molar	0.9943	0.9943	0.9943	0.9914
Premolar	0.9944	0.9944	0.9944	0.9917
Cap	0.9970	1.0000	0.9985	0.9900
Impacted	0.9941	0.9912	0.9927	0.9912
Average	0.9950	0.9950	0.9941	0.9931

TABLE 3. Confusion matrix.

Class	Molar	Premolar	Canine	Incisors	Cap	Impacted
Molar	347	1	1	0	1	0
Premolar	1	357	1	0	0	1
Canine	0	1	298	0	0	1
Incisors	0	0	0	324	1	0
Cap	0	0	0	0	325	0
Impacted	0	0	1	1	1	337

• Confusion Matrix

A confusion matrix is used to evaluate the performance of the classification model by comparing the predicted labels with the true labels. It shows the counts of true positives, true negatives, false positives, and false negatives, helping assess the model's accuracy, sensitivity, and specificity.

After training our model, Table 2 summarizes the achieved metrics: Precision, Recall, and F1 score all reached 99%, with an overall accuracy of 99.31%. The confusion matrix for the model is presented in Table 3. To further validate the model's generalizability, we tested it on an additional external datasets sourced from Mendeley [38]. The performance on this external dataset was also highly favorable, with Precision at 99.2%, Recall at 99.1%, F1-Score of 99.15%, and Detection and Classification Accuracy of 99.05%. These results highlight the effectiveness of our approach in achieving high accuracy and reliability of dental tooth classification using OPG X-ray images.

E. DISCUSSION

Deep learning has shown significant promise in the exploration of advanced techniques for dental imaging analysis. Muramatsu et al. [28] developed a CNN that applies cross-validation to classify various tooth types and detect anomalies. This study included incisors, canines, premolars, molars, non-metal restored, partially restored, and completely restored teeth. The proposed approach achieved classification accuracies of 93.2% and 98.0% for tooth types and anomalies respectively. Small datasets affected the robustness of the system. In the field of dental disease analysis, the Curvilinear Semantic Deep Convolutional Neural Network (CSDCNN) alongside Inception ResNetV2 was introduced in [29] for the segmentation and classification of various tooth diseases, including dental caries, periapical infection, periodontal disease, and pericoronal disease. This approach attained an average accuracy of 94.51%. The model is susceptible to noise, which affects its accuracy.

To enhance dental diagnostics, an innovative method for identifying and categorizing dental types and abnormalities was proposed in [30]. The dropout method has been used to improve the network classification performance and reduce overfitting. This model faces difficulties in detecting small pathologies and potentially struggling with subtle or minor dental abnormalities.

In the realm of 3D dental imaging, 3D Dental Image Segmentation and Classification using DL with Tunicate Swarm Algorithm (3DDISC-DLTSA) model was introduced in [31] to segment and classify seven distinct tooth types. This approach involves image filtering, image preprocessing, U-Net segmentation, and the DenseNet-169 model for feature extraction. It achieved accuracies of 96.67% and 97.35% for different datasets. The hyper-parameter limitations in U-Net for tooth segmentation accuracy remain a challenge.

In exploring advanced methodologies for dental image classification, an approach based on a CNN with a dropout regularization methodology was suggested in [32] to classify two types of teeth and two distinct defects. In this study the fully linked layer of the CNN was trained using hybrid GA-BP learning. In the domain of dental image segmentation, a transformer-based U-Net architecture featuring skip connections, was presented in [33] and was designed for segmenting teeth in panoramic radiographs. The study compared this architecture with other baselines such as U-Net, Link-Net, and FPN, demonstrating superior performance over these methods.

In advancing techniques for tooth segmentation and caries detection, an ensemble transfer learning approach using a capsule network was presented in [34] for tooth segmentation and caries detection in panoramic X-rays. Their method achieved recall scores of 69.44% and 90.52% for mild and severe carious lesions, respectively. Concerns regarding potential overfitting due to the small dataset size may have influenced the model's performance. To improve dental image classification, the YOLOv3 model was employed in [35] to classify impacted molars, achieving a classification accuracy of 96%. YOLOv3 is less effective at capturing global spatial features.

A comparison of the proposed method with other state of the art methods is shown in table 4. Upon reviewing previous related studies, our results indicate that the Transformer-Based YOLO framework significantly surpasses the current methods in dental pathology detection and tooth type categorization, achieving an impressive accuracy of 99.31%. Unlike previous studies, our approach, which incorporates multi-head self-attention and Genetic Algorithm optimization, yields higher precision, recall, and F1 scores. By addressing challenges such as dataset size limitations and noise sensitivity, this methodology demonstrates promising potential for the accurate and reliable detection, segmentation, and classification of diverse tooth types and anomalies in dental radiography. This highlights the importance of integrating sophisticated deep learning architectures and optimizing techniques to improve the accuracy of dental diagnostics.

TABLE 4. Comparison of the proposed method with other state-of-the-art models.

Authors	Year	Model	Modality	Tooth Types and Anomalies	Classification Accuracy	Limitations
Oktay [9]	2017	Modified AlexNet CNN	Panoramic X-ray (OPG)	Tooth Types	Accuracy: over 90%	Challenges with misalignment in neighboring teeth
Miki et al. [10]	2017	AlexNet with manual ROI	Dental CT	Tooth Types	Accuracy: 88.8%	Preprocessing variability and metal artifact challenges
Raith et al. [11]	2017	ANN	3D Tooth Cusps	Tooth Types	Accuracy: 93.5%	Misclassification dependency on training data size
Wirtz et al. [13]	2018	Neural Network + Shape Model	Panoramic X-ray (OPG)	Tooth Types	Precision: 0.790; Recall: 0.827	Excludes wisdom teeth; challenges in unfamiliar regions
Zhang et al. [14]	2018	3D CNN	Dental periapical X-ray	Tooth Types	Precision: 95.8%; Recall: 96.1%	Requires substantial datasets for optimal performance
Xu et al. [15]	2018	Deep CNN	3D Dental Mesh Models	Tooth Types	Mesh labeling accuracy: 99.06%	Confinement to dental mesh segmentation
Li et al. [12]	2019	Deep CNN with Global Average Pooling	Cone-beam CT	Tooth Types	Average sensitivity: 87%	Noise affects accuracy
Tian et al. [16]	2019	Sparse Voxel Octree (SVO) + CNN	3D Dental Models	Tooth Types	Classification: 95.96%; Segmentation: 88.06%	Limited data; hierarchical segmentation impacts accuracy
Anuradha et al. [17]	2020	Faster R-CNN	Panoramic X-ray (OPG)	Teeth Types and Anomalies	Detection: 90%; Classification: 99%	Misclassification of adjacent teeth
Singh et al. [21]	2020	Deep CNN	Panoramic X-ray (OPG)	Tooth Types	Accuracy: 95%	Susceptible to misclassification of adjacent teeth
Dascalu et al. [36]	2023	Faster R-CNN	Panoramic X-ray (OPG)	Teeth Types and Anomalies	Precision: 0.49; F1 scores: 0.71 (Healthy teeth), 0.76 (Multi-label)	Small dataset limits generalizability; needs broader evaluation
Park et al. [37]	2023	Deep Learning Algorithm	panoramic and periapical radiographic images	Teeth Anomalies	Accuracy: 88.53%; Precision: 85.70%; Recall: 82.30%	Needs diverse datasets for generalizability; bias in image quality and demographic factors
Walid et al. [39]	2023	Mask RCNN	Panoramic X-ray (OPG)	Teeth Types	Precision: 96%	The model's performance may be improved
Abraham et al. [20]	2024	U-Net, YOLOv3	Panoramic X-ray (OPG)	Teeth Anomalies	Accuracy: 80%	Accuracy may be improved
Proposed Method	2025	Transformer, YOLO	Panoramic X-ray (OPG)	Teeth Types and Anomalies	Accuracy: 99.31%	Performance on low-quality imaging may be low

F. LIMITATION

Despite achieving a high detection and classification accuracy, the proposed Transformer-based YOLO with Genetic Algorithm (GA) optimization has certain limitations. The model's performance is highly dependent on the quality and diversity of the training dataset, which may not adequately cover all variations in dental anomalies or imaging conditions. The robustness of the model under noisy or low-quality imaging conditions and its applicability to other imaging modalities, such as cone-beam CT or 3D scans, remain unvalidated.

VI. CONCLUSION

Our work presents a novel method for automating the recognition of dental pathology and classification of tooth types from panoramic X-ray images by utilizing a Transformer-Based YOLO architecture. The CSP-Darknet basis for YOLOv4 also incorporates a multi-head self-attention block, which improves the model ability to extract complex feature representations and gather rich contextual

information from dental images. Our method greatly increases the image quality and analysis accuracy when combined with various preprocessing techniques such as image resizing, rotation, Gaussian smoothing, and histogram scaling. Using Genetic Algorithm (GA) optimization for hyperparameter tuning, the model parameters are further optimized, which adds to its exceptional performance metrics, including an average precision, recall rate, and F1 score of over 99.00%. Furthermore, the system attained an impressive detection and classification rate of 99.31%. The objective of our proposed system is to improve dental healthcare practices by reducing errors, expediting the diagnostic process, and offering a reliable solution.

VII. FUTURE SCOPE

Expanding the dataset to include diverse imaging modalities, patient demographics, and rare dental anomalies can enhance model generalizability. Additionally, integrating noise-resilient preprocessing techniques and advanced image augmentation strategies can improve performance in

challenging imaging scenarios. Incorporating explainable AI (XAI) methods will provide interpretability to aid clinical decision-making. Further investigation into adapting the framework for 3D dental imaging and cone-beam CT analysis holds significant potential for advancing automated dental diagnostics.

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DECLARATION

Informed consent was obtained from all individual participants included in the study. All OPG images were anonymized to ensure confidentiality.

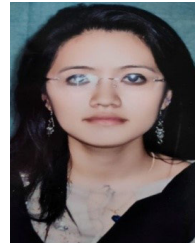
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